

# MD FAHIM UL ISLAM

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## RESEARCH INTERESTS

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- Designing interpretable multimodal generative models in semiconductor manufacturing challenges.
- Machine learning for hardware and physical design optimization.

## EDUCATION

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### Doctor of Philosophy (PhD) in Computer Science

*Fall 2025 – Ongoing*

*Arizona State University (ASU), Tempe, Arizona, USA*

*Supervisor: Prof. Krishnendu Chakrabarty*

*Focus: AI/ML for hardware design optimization; Generative AI for semiconductor manufacturing*

### Bachelor of Computer Science and Engineering

*Spring 2019 – Spring 2023*

*Brac University, Dhaka, Bangladesh*

## EMPLOYMENT EXPERIENCE

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- **Graduate Research Associate**, *School of ECEE, Arizona State University* *Tempe, AZ, 2025 – Present*
  - Developing multi-agent AI systems for automated root cause analysis of semiconductor wafer defects.
  - Building scalable ML pipelines for wafer defect classification using vision-language models.
  - Collaborating with DFT research groups on reliability-aware defect detection and test data analytics.
- **Graduate Research Assistant**, *Dept of CSE, Brac University* *Dhaka, Bangladesh, 2023 – 2025*
  - Developed federated learning systems with robust aggregation and defense mechanisms (Krum, Flame, FLPruning) for secure distributed model training.
- **ML Engineer Intern**, *Virtual Market Solution Limited (VMSL)* *Dhaka, Bangladesh, 2023*
  - Implemented production ML models for data-driven decision support; automated data preprocessing Python.

## SELECTED PUBLICATIONS

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- **DefectVICL: Data-Efficient Wafer Defect Classification with Vision In Context Learning (Accepted)**  
Md Fahim Ul Islam, Soyed Tuhin Ahmed, John Carulli, Krishnendu Chakrabarty  
*44th IEEE VLSI Test Symposium*, 2026
- **SOLDER: State-Space Oriented Learning with Denoising-Enhanced Representations for PCB Defect Analysis (Accepted as Poster)**  
Md Fahim Ul Islam, Krishnendu Chakrabarty  
*European Test Symposium (ETS)*, 2026
- **A Resource-Efficient Federated Learning Framework for Intrusion Detection in IoMT Networks**  
Md Fahim Ul Islam, Amitabha Chakrabarty, and Md. Golam Rabiul Alam, Siti Sarah Maidin  
*IEEE Transactions on Consumer Electronics* (Impact Factor: 10.3)
- **FedWGCA: A Federated Learning Based UAV Intrusion Detection with Gradient Clipping and Attention-Based Neural Networks**  
Md Fahim Ul Islam, Amitabha Chakrabarty, Halimatun Saadiah Hakimi, Siti Sarah Maidin  
*IEEE Open Journal of the Computer Society* (Impact Factor: 8.2)
- **Optimizing Patient Feedback with Generative Adversarial Network Leveraging Knowledge Distillation to Improve Healthcare**  
Md Fahim Ul Islam, Amitabha Chakrabarty, Abrar Hasan Efaz, Xiaoding Wang, Md. Jalil Piran  
*IEEE Journal of Biomedical and Health Informatics* (Impact Factor: 6.7)

## RELEVANT PROJECTS

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### Automated Wafer Defect Classification Pipeline

- Built end-to-end data processing and evaluation pipelines for wafer map datasets, implementing CLIP-based similarity search, centroid/diversity-based example selection, and novel faithfulness metrics.

### Federated Learning with Backdoor Attack/Defense Mechanisms

- Implemented a modular federated learning framework with multiple backdoor attack vectors .

### Optimizing Patient Feedback Using GANs with Knowledge Distillation

- Designed a transformer-based GAN with knowledge distillation to analyze patient reviews from private hospitals.

### Explainable Multi-Modal DeepFake Detection Using AV-HuBERT

- Developed an explainable DeepFake detection framework using AV-HuBERT on audio-visual datasets.

## AWARDS

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- Graduate Fellowship awarded by Arizona State University for outstanding academic performance.
- Won 1<sup>st</sup> position in a cybersecurity quiz competition sponsored by Grameenphone Academy.
- Reached finalist as 9<sup>th</sup> position in NLP national hackathon sponsored by BdOSN.
- Achieved 85% Tution Fee Waiver Scholarship for the Master's program by Brac University.
- Achieved 8<sup>th</sup> position of the Hult Prize competition sponsored by Brac University.

## TECHNICAL SKILLS

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**Programming:** Python, Java, Javascript, PHP

**Virtualization:** VMware, VirtualBox

**Libraries :** NumPy, Pandas, Matplotlib

**Visulazation :** PowerBI, Microsoft Powerpoint

**Writing:** L<sup>A</sup>T<sub>E</sub>X, Microsoft Word, Microsoft Excel, Notion

**Frameworks:** TensorFlow, PyTorch, PyCharm, OpenGL, Flask, Flutter, Unity

**Simulation:** LTSPICE, Proteus, Cisco Packet Tracer

**Hardware** Raspberry Pi, Arduino

## REVIEWER EXPERIENCE

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- **Journal of Electronic Testing**, Springer 2026
- **Engineering Applications of Artificial Intelligence (EAAI)**, Elsevier 2025
- **IEEE Access**, IEEE 2024
- **Scientific Reports**, Nature Portfolio 2024

## TEACHING & DEPARTMENTAL SERVICES

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- **Undergraduate Thesis Panel Judge**, *Dept of CSE, Brac University* 2023 - 2025
- **Undergraduate Thesis Co-supervisor**, *Dept of CSE, Brac University* 2023 - 2025

## INVITED SEMINAR TALKS

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- **Keynote Speaker**, “Generative AI: From Concepts to Research,” *IEEE CS BRACU SBC Webinar* 2024
- **Guest Speaker**, “Exploring Generative AI,” *Grad Bunker Academy Webinar* 2024